

A New Approach to the Treatment of Nasal Bone Fracture: Radiologic Classification of Nasal Bone Fractures and Its Clinical Application

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Purpose: A radiologic examination is required in the treatment of nasal bone fracture to determine the fracture condition. Thus, there is an increasing need for radiologic classification of nasal bone fractures that can be applied to clinical practice.

Materials and Methods: Computed tomography was performed in 125 patients with nasal bone fractures to determine which axial view best showed the entire nasal view. The obtained axial view was then used as a reference for classification. The length from the top to the base of the nasal bone was divided into upper, middle, and lower levels, after which the fracture location was determined. If the fracture spanned the boundaries of these levels, it was classified as the total level. Subsequently, the fracture was subclassified based on the fracture direction and pattern and the concurrent fracture.

Results: Radiologic examination of patients with nasal bone fracture showed that nasal bone fracture was frequently found at the total, middle, upper, and lower levels, in that order. Nasal bone fractures at the upper level showed lower frequencies of complication and reoperation than the fractures at the other levels, whereas nasal bone fractures at the total level showed the highest frequencies of complication and reoperation.

Conclusion: Radiologic classification can be useful for preoperative and postoperative evaluations of nasal bone fractures and can be helpful in understanding such fractures because it can efficiently predict the prognosis of a fracture.

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Nasal bone fractures account for the largest proportion of facial traumas. The number of cases of nasal bone fracture has been increasing with the increase in the number of injuries or car accidents.¹ Nasal bone fracture can lead to complications such as a severely deformed nose and intranasal dysfunction unless it is treated. Furthermore, if it is not treated properly, the morbidity of complications will be higher than with other facial fractures.²

To properly treat nasal bone fracture, it is important to diagnose and evaluate it by a physical or

radiologic examination. In general, nasal bone fracture can be diagnosed by identifying crepitus and tenderness in a physical examination. A radiologic examination is commonly conducted to check the nasal bone, the fractured bone, and the nasal septum. With the development of radiology and related technologies, most nasal bone fractures can be diagnosed with radiographs and with computed tomography (CT). Radiography is also required for accurate treatment and postoperative evaluation, in addition to the diagnosis of nasal bone fracture.³ A radiologic classification of nasal bone fractures, which plays a key role in the diagnosis and evaluation of nasal bone fractures, has not been established, and its necessity has been increasing.

The present investigators classified nasal bone fractures based on radiography, which is commonly used before and after an operation, and further investigated the characteristics and prognoses of nasal bone fractures based on fracture location according to a radiologic classification. Accordingly, this study reports the

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usefulness of a radiologic classification in understanding nasal bone fractures.

Materials and Methods

Radiographs and CT, which produce a nasal lateral view and a Water view, are commonly conducted to diagnose nasal bone fractures. In general, if abnormal findings are seen on plain x-ray, CT is conducted. These 2 radiologies are also commonly conducted to identify other facial fractures in patients with facial fractures. Therefore, if nasal bone fractures can be classified based on the 2 radiologic examinations, then such examinations will be useful in clinical practice.

The nasal lateral view, Water view, and computed tomographic facial view were taken to analyze nasal bone fractures of 125 patients who visited the Kosin University Gaspel Hospital from January 2006 through December 2008. Before the study, the researchers obtained informed consent from all patients and institutional review board approval from the clinical and educational institutions. Plain x-ray and CT were conducted in all patients before an operation, followed by closed reduction to treat the nasal bone fracture. A nasal lateral view and a Water view were taken 1 day after the operation. Plain x-ray was conducted 1 more time 1 week after the operation while the nasal pack was removed. CT was also conducted to evaluate concurrent fractures depending on its necessity, but it was not a routine study. Subsequently, plain x-ray was conducted for 6 months at a 2-month interval to observe complications and the need for additional treatment such as corrective rhinoplasty because of complications.

For radiologic classification of nasal bone fractures, the computed tomographic axial view that best showed the entire nasal bone was used as a reference for classification.⁴ The length from the top to the base of the nasal bone was divided into upper, middle, and lower levels, after which the fracture location was determined. If the fracture spanned the boundaries of these levels, it was separately classified as the total level (Figs 1, 2). The fracture location was evaluated horizontally, after which the right side, left side, and right and left sides were denoted as R, L, and B, respectively. In addition, according to the pattern of the fracture, ie, depressed, elevated, cracked (greenstick), and comminuted, the fractured bone was denoted as D, E, G, and C, respectively. In principle, if a fracture was seen on the right and left sides, the right side was described first. No concurrent fracture, nasal septal fracture, another concurrent fracture, and nasal septum fracture and another concurrent fracture were denoted as N, S, O, and SO, respectively (Table 1).

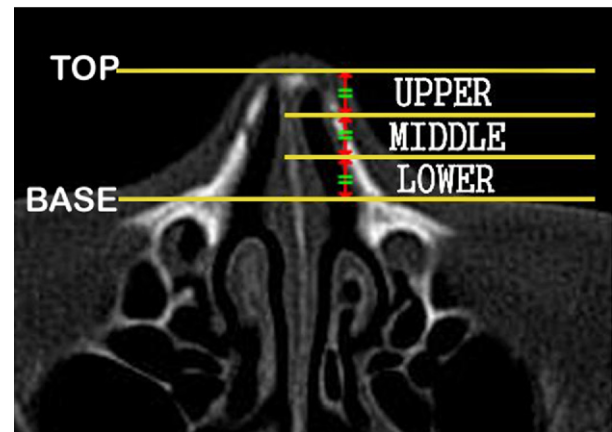


FIGURE 1. Based on a computed tomographic axial view, the length from the top to the base of the nasal bone was divided into upper, middle, and lower levels.

Han, Han, and Park. *New Approach to Nasal Bone Fracture. J Oral Maxillofac Surg* 2011.

In the radiologic classification that was applied clinically, if the fractured nasal bone without a concurrent fracture was depressed at the left side of the upper level, it was denoted as upper-level L-D-N; if the fractured nasal bone was accompanied by a maxillary segmental fracture and the septal fracture was depressed to the right side of the middle level, it was denoted as middle-level R-D-SO. If the fractured nasal bone that was accompanied by a blowout fracture was depressed to the right side of the lower level, it was denoted as lower-level R-D-O; if the fractured nasal bone without another fracture was depressed to the right side of the lower level and was elevated to the left side of the middle level, it was denoted as total-level B-DE-N (Fig 2). Based on these denotations, all nasal bone fractures were easily denoted to review the information that should be reviewed before the operation.

According to this radiologic classification, the distribution of nasal bone fracture locations, complications, and reoperation rate were investigated retrospectively. In addition, the direction and form of the fractured bone based on fracture location and concurrent fractures were identified, and the morbidity of complications and reoperation rate based on each characteristic were investigated. Complications such as a fine imbalance of the nasal bone, a hump nose, and septal deviation in patients with nasal bone fracture were classified as deformities of the external nose, whereas complications such as intranasal synechia and nasal obstruction caused by nasal septal deviation were classified as intranasal dysfunctions. The percentages that were used in this study were rounded off and recorded to the nearest tenth.

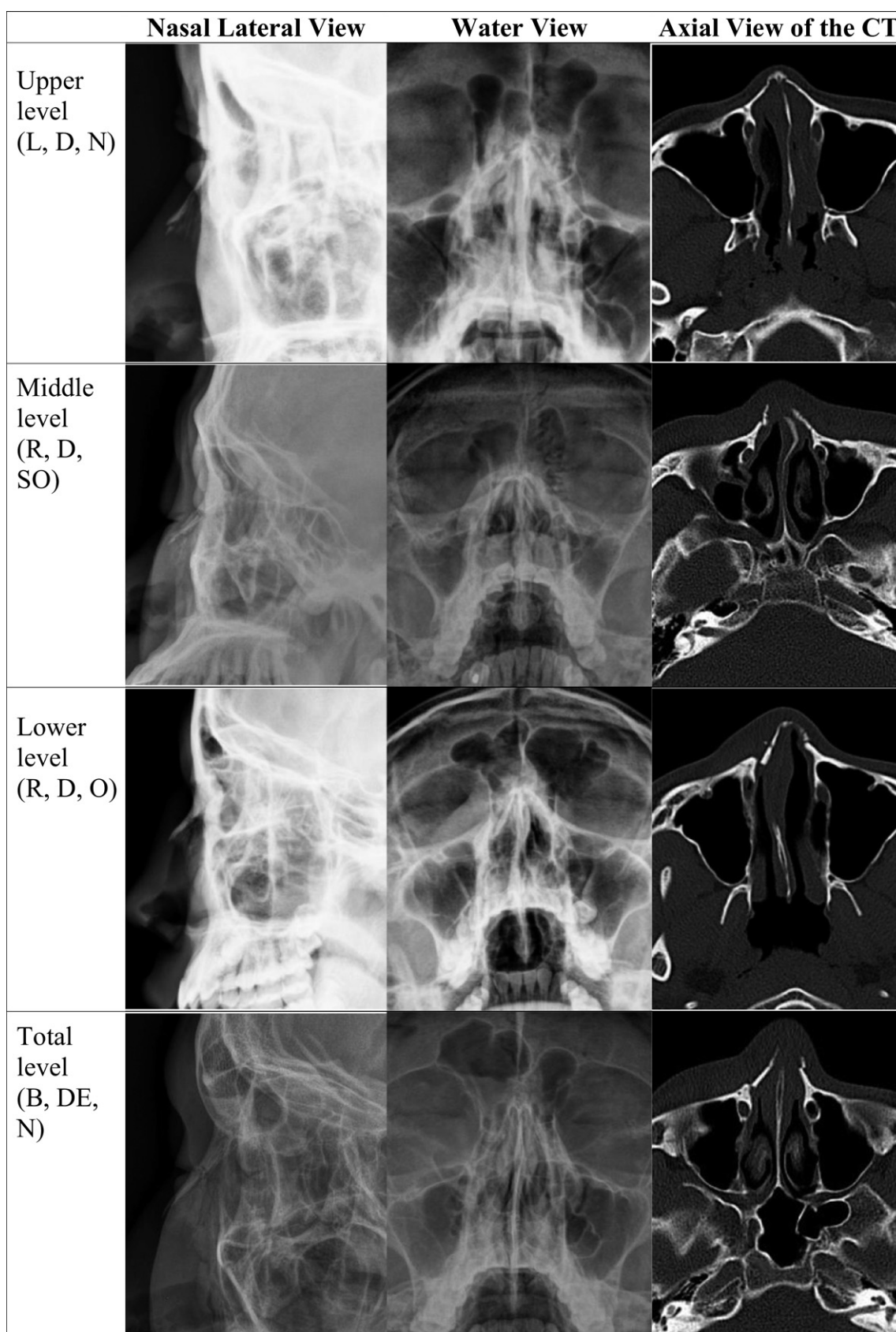


FIGURE 2. Radiographic classifications of the nasal bone. B, right and left; C, comminution; D, depression; E, elevation; G, greenstick; L, left; N, none; O, other combined; R, right; S, septum.

Han, Han, and Park. New Approach to Nasal Bone Fracture. J Oral Maxillofac Surg 2011.

Table 1. DETAILED CLASSIFICATION AND DISPLAY PATTERNS OF THE NASAL BONE

Detailed Classification	Display Pattern
Fracture site	B, left and right; L, left; R, right
Fracture pattern	C, comminution; D, depression; E, elevation; G, greenstick
Combined fracture	N, none; O, other combined; S, septum

Han, Han, and Park. *New Approach to Nasal Bone Fracture. J Oral Maxillofac Surg* 2011.

Table 3. FRACTURE DIRECTION BASED ON NASAL BONE FRACTURE LOCATION

	Right (%)	Left (%)	Right and Left (%)
Upper level	12 (42.9)	11 (39.3)	5 (17.9)
Middle level	9 (25.7)	16 (45.7)	10 (28.6)
Lower level	12 (48.0)	10 (40.0)	3 (12.0)
Total level	4 (10.8)	5 (13.5)	28 (75.7)
Total	37 (29.6)	42 (33.6)	46 (36.8)

Han, Han, and Park. *New Approach to Nasal Bone Fracture. J Oral Maxillofac Surg* 2011.

Results

Radiologic examination of patients with nasal bone fracture showed that the fracture was frequently found at the total, middle, upper, and lower levels, in that order. Because a fracture at the lower level, which is the thickest portion of the nasal bone, occurs from a strong impact, it was commonly classified under the total level because it occurs in the middle and upper levels and is not restricted to the lower level. Of 125 patients, 18.4% complained of external nose deformity and 15.2% complained of intranasal dysfunction. The reoperation rate was 5.6%.

Nasal bone fractures at the upper level showed lower frequencies of complications and reoperations than fractures at the other levels. In particular, the frequency of complications related to intranasal dysfunction was significantly lower. In contrast, nasal bone fractures at the total level showed the highest frequencies of complications and reoperations (Table 2). For the direction of nasal bone fractures, fractures at the right and left sides accounted for 36.8%, which was the highest frequency, and fractures at the left side and at the right side accounted for 33.6% and 29.6%, respectively. For fractures at the middle level, the frequency of fractures at the left side was higher. The frequency of fractures at the right and left sides was significantly lower at the lower level and significantly higher at the total level (Table 3). For the pattern of nasal bone fractures, fractures at the

right and left sides were classified individually. Depression accounted for 45%, which was the highest frequency. For patterns of nasal bone fractures based on location, greenstick accounted for 39.4%, which was the highest frequency at the upper level, whereas comminution accounted for 6.1%, which was lower at the upper level. Comminution accounted for 35.4%, which was the highest frequency at the total level (Table 4). Nasal bone fractures alone accounted for 52%, concurrent nasal septal fractures accounted for 30.4%, and other concurrent facial fractures such as at the orbital wall or maxilla accounted for 24%. For fractures restricted to the upper level, because these could occur from a weak impact, nasal bone fractures alone accounted for the largest proportion. In contrast, nasal bone fractures were relatively less frequently accompanied by nasal septal or other facial fractures. Nasal septal fractures or other facial fractures more frequently accompanied the nasal bone fractures at the total level (Table 5). For a prognosis based on characteristics of nasal bone fracture, morbidity from complications was higher in cases of nasal bone fracture at the right and left sides. Morbidity from complications was lower for greenstick fractures, but frequencies of complications and reoperations were higher for comminuted fractures. In addition, frequency of complications was higher in cases with concurrent nasal septal fractures. In particular, frequency of complications related to intranasal dysfunctions because of nasal

Table 2. DISTRIBUTION AND POSTOPERATIVE PROGNOSIS OF NASAL BONE FRACTURE BASED ON ITS LOCATION

	No. of Patients	Nasal Deformity (%)	Dysfunction of Nasal Cavity (%)	Reoperation (%)
Upper level	28	4 (14.3)	2 (7.1)	1 (3.6)
Middle level	35	7 (20.0)	6 (17.1)	2 (5.7)
Lower level	25	3 (15.6)	4 (16.0)	1 (4.0)
Total level	37	9 (24.3)	7 (18.9)	3 (8.1)
Total	125	23 (18.4)	19 (15.2)	7 (5.6)

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Table 4. FRACTURE PATTERN BASED ON NASAL BONE FRACTURE LOCATION

	Depression (%)	Elevation (%)	Greenstick (%)	Comminution (%)	Total
Upper level	12 (36.7)	6 (18.2)	13 (39.4)	2 (6.1)	33
Middle level	28 (62.2)	7 (15.6)	3 (6.7)	7 (15.6)	45
Lower level	20 (71.4)	3 (10.7)	2 (7.1)	3 (10.7)	28
Total level	17 (26.2)	19 (29.2)	6 (9.2)	23 (35.4)	65
Total	77 (45.0)	35 (20.5)	24 (14.0)	35 (20.5)	171

Han, Han, and Park. *New Approach to Nasal Bone Fracture. J Oral Maxillofac Surg* 2011.

septal deviation was higher and led to an increased reoperation rate (Table 6).

Discussion

In 1978, Stranc and Robertson⁵ classified detailed fracture patterns based on the direction of the impact and its concurrent damage. According to the impact direction, impacts were classified as frontal impacts and lateral impacts. According to the degree of damage to the nasal bone and nasal septum, the impact was further classified into types 1, 2, and 3. This classification system is most commonly used for nasal bone fractures, but it has some limitations in its application to clinical practice. Most nasal bone fractures are evaluated not by gross examination but by radiologic examination, and various classifications of impact direction and degree of damage cannot satisfy diverse treatments in clinical practice. That is, except for type 3 frontal impact and type 3 lateral impact, there is no difference in treatment methods for diverse fracture types. Therefore, this classification system is not clinically useful for the treatment of nasal bone fractures.

Murray et al⁶ examined nasal bone fractures in 50 corpses and classified these fractures as types 1 to 7. The classification criteria, however, were based on the nasal septum, and the classification was conducted in corpses with a mean age of 71.6 years. Because nasal septal fracture frequently occurs from

the teenage years through the 30s (especially in the 20s), the strength of the bone and cartilage of the corpses was likely to have differed from that of common nasal septal fractures. In addition, because the bone and cartilage could have been deformed by chemicals used in the corpses, it is difficult to consider this classification as a general classification of nasal bone fractures.

Manson et al⁷ used CT to classify facial fractures into low-energy, middle-energy, and high-energy fractures based on the degree of exodeviation and posterior deviation of the nasal bone fracture and bilateral nasal bone fracture. The direction of the external impact that caused the nasal bone fracture was classified as antero-posterior and lateral directions. Fracture patterns differed, however, because more impact can be exerted diagonally to the bone; in addition, prediction of the fracture pattern based on impact direction is only estimation. It is difficult to fully identify other concurrent fractures using this classification.

Before radiologic classification of nasal bone fractures, the present investigators summarized the identification of the criteria for nasal bone fracture and evaluation conducted by clinicians.

1. Determine the location of the nasal bone fracture—the side wall, the dorsum of the nasal bone, or the entire nasal bone.
2. Determine if the fracture is located at the right, the left, or bilaterally.
3. Determine if the fractured bone has been medially displaced or laterally displaced and identify whether the fracture is a slight crack or severe comminution.
4. Determine if there are other concurrent fractures, including nasal septal fracture. Open reduction or closed reduction can be chosen because of the identification. In general, if severe nasal septal fracture or another facial fracture accompanies the nasal bone fracture, open reduction is commonly conducted, whereas closed reduction is conducted for mild nasal septal fracture.

When a new classification of nasal bone fractures was initiated, the present investigators maintained

Table 5. CONCURRENT FRACTURE BASED ON NASAL BONE FRACTURE LOCATION

	None (%)	Septum (%)*	Other Combined (%)*
Upper level	21 (75.0)	4 (14.3)	3 (10.7)
Middle level	17 (48.6)	14 (40.0)	7 (20.0)
Lower level	12 (48.0)	8 (32.0)	7 (28.0)
Total level	15 (40.5)	12 (32.4)	13 (35.1)
Total	65 (52.0)	38 (30.4)	30 (24.0)

*Nasal septal fracture and other facial fractures were seen in 8 patients.

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Table 6. DISTRIBUTION AND POSTOPERATIVE PROGNOSIS BASED ON CHARACTERISTICS OF NASAL BONE FRACTURE

	No. of Patients (%)	Nasal Deformity (%)	Dysfunction of Nasal Cavity (%)	Reoperation (%)
Fracture site				
Right	37 (29.6)	4 (10.8)	2 (5.4)	1 (2.7)
Left	42 (33.4)	5 (11.9)	3 (7.1)	1 (2.4)
Right and left	46 (36.8)	14 (30.4)	14 (30.4)	5 (10.9)
Fracture pattern				
Depression	77 (45.0)	5 (6.5)	3 (3.9)	2 (2.6)
Elevation	35 (20.5)	3 (8.6)	1 (2.9)	1 (2.9)
Greenstick	24 (14.0)	0 (0.0)	1 (4.2)	0 (0.0)
Comminution	35 (20.5)	15 (42.9)	14 (40.0)	4 (11.4)
Combined fracture				
None	65 (52.0)	6 (9.4)	3 (4.6)	1 (1.5)
Septum	38 (30.4)	11 (28.9)	14 (36.8)	4 (10.5)
Other combined	30 (24.0)	6 (20.0)	2 (6.7)	2 (6.7)

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that the new classification should be clinically useful for the evaluation and treatment of nasal bone fractures. Therefore, it was believed that a classification system that could provide useful information for the treatment of nasal bone fractures was more valuable than a classification system that could identify all types of nasal bone fracture. In reality, to diagnose and treat patients with nasal bone fracture in clinical practice, accurate identification of fracture location, direction and pattern of the fractured bone, and concurrent fractures are more important than determination of the degree of damage based on impact direction or the power to increase a patient's satisfaction and decrease complications. Based on this principle, a new classification was made using radiologic examination, which is commonly conducted before and after an operation, to evaluate the nasal bone. Furthermore, criteria for the denotation and classification included factors that are simple and can easily explain nasal bone fracture characteristics that should be considered in clinical practice. In addition, the classification criteria were not determined for information that is seldom useful for the treatment of nasal bone fracture.

As mentioned in the Results, nasal bone fracture was frequently found at the total, middle, upper, and lower levels, in that order. This result cannot fully explain, however, the frequency of nasal bone fractures based on their location. This is because this study was conducted in patients with nasal bone fracture who visited the investigators' hospital as urgent inpatients or outpatients because of trauma. That is, patients who were not aware that they had nasal bone fracture, such as greenstick at the upper level, even if they had nasal bone fracture, or patients who did not wish to be treated even if they were aware that they had nasal bone fracture were not included as

study subjects. This shortcoming is likely to be negligible because the investigators' classification of nasal bone fractures focused more on providing information for the effective treatment of patients with nasal bone fracture than on the classification of fractures.

After the evaluation of the postoperative condition of patients with nasal bone fracture according to the present radiologic classification of nasal bone fractures, the following results were obtained. Fractures at the upper level of the nasal bone that were restricted to the nasal bone's dorsum commonly occurred alone rather than being accompanied by other fractures because of the weak impact that caused them. Particularly, because nasal septal fracture is rarely accompanied by other fractures, it produces few complications such as intranasal dysfunction. In contrast, fractures at the middle level, which is the weakest area of the nasal bone, had the highest frequency of occurrence and occurred more frequently at the left side of the nasal bone because the impact that caused the fracture was frequently from the right punch of attackers. In addition, because a nasal septal fracture is commonly accompanied by another fracture, it is important to prevent related complications. Fractures at the lower level of the nasal bone had more depression patterns than other fractures. These rarely occurred at the right and left sides and were commonly accompanied by other facial fractures. Fractures at the total level of the nasal bone covered large areas because of strong impact. Thus, these commonly occurred at the right and left sides rather than unilaterally. Because such fractures show a high frequency of comminution and concurrent nasal septal or other facial fractures, these also produce a high frequency of complications or reoperation. Based on these findings, results of an evaluation and prognosis of patients can be easily predicted based on the loca-

tion of the nasal bone fracture according to the fracture's radiologic classification. In addition, matters that require attention can be reviewed during preparations for the operation.

When nasal bone fracture is analyzed with radiologic examination, there is a difference in the information that can be obtained from the nasal lateral view and the Water view, depending on the fracture location. That is, on plain x-ray, for a dorsum fracture, which is at the upper level, the fracture pattern can be observed well in the nasal lateral view and rarely observed in the Water view (Fig 2). Conversely, for a sidewall fracture, which is at the middle level and lower level, the fracture pattern can be observed well in the Water view and rarely observed in the nasal lateral view (Fig 2). The cost of treatment can likely be decreased because the necessary radiologic examination can be conducted before and after the operation based on the aforementioned findings.

For a fracture that is at the boundary between the dorsum and the sidewall of the nasal bone at the middle level, the fracture pattern can be clearly seen in the computed tomographic axial view, whereas it might not be observed at all in the nasal lateral or Water view. Therefore, evaluation of nasal bone fracture with only a radiologic examination may cause errors that would result in an inaccurate diagnosis. Therefore, in addition to a radiologic examination, a physical examination should be conducted for the diagnosis and evaluation of nasal bone fracture.

The most important goal of treatment of patients with nasal bone fracture is to return the patient's condition to his/her condition before the injury.⁸ To achieve this goal, it is important to conduct a radiologic examination from which the location of the nasal bone fracture can be clearly determined because an accurate evaluation of the nasal bone fracture should be conducted before the operation. In addition, results of the postoperative evaluation

should be compared with results of the preoperative radiologic examination. Therefore, it is likely that radiologic classification of nasal bone fractures has clinical usefulness.

According to the radiologic classification of nasal bone fractures, clinical information such as location of the nasal bone fracture, fracture pattern, and concurrent fractures can be easily analyzed. Furthermore, the characteristics and prognosis of nasal bone fractures based on fracture location can be predicted by the radiologic classification. In addition, because nasal bone fractures can be summarized with a simple denotation according to this classification system, the appropriate operation method can be determined, and matters that require attention during the operation can be reviewed. Therefore, if the present radiologic classification of nasal bone fractures is applied in clinical practice, it will be useful in decreasing complications because important factors of the treatment of nasal bone fracture can be identified.

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